

PFPA Summer Intern 2026 — Team 1

Weekly Report — Week 2 (ending June 26, 2026)

Project: AI-Based Credit Rating Framework (KMV / TiC)

Team Lead: Shenyuan Wu

Team Members: Guanming Lin, Haixing Tan

1. Summary

This week, all three team members independently built automated data pipelines to collect the market and financial data required for the KMV credit rating model. Each member approached the problem from a different angle — Yahoo Finance Chart API + SEC EDGAR (Shenyuan), SEC EDGAR XBRL + Yahoo Finance + FRED with Excel output (Haixing), and yfinance + FRED with CSV output (Guanming). Together, the team established a validated, reproducible dataset covering all 10 target companies across three data categories: stock market data, company financial data, and macro interest rate data.

Target companies: COST, KO, DELL, ORCL, PNC, WMT, INTU, AMZN, T, KHC

2. Work Completed

2.1 Shenyuan Wu — Production Notebook (`Week2 - Updated Data Downloading`)

Built and published (`PFPA_Week2_Data_Download_v3.ipynb`), a fully automated Jupyter notebook replacing the Week 1 (`yfinance`)-based prototype. Key design decision: eliminate (`yfinance`) entirely to avoid Yahoo Finance rate limits that blocked all 10 tickers simultaneously in Week 1.

Data sources used:

- **Stock prices + dividends:** Yahoo Finance Chart API via direct HTTP (`/v8/finance/chart/{ticker}`) — no library dependency
- **Balance sheet / financials:** SEC EDGAR XBRL API (`data.sec.gov/api/xbrl/companyfacts/CIK{cik}.json`) — 2 most recent 10-K filings per company
- **Shares outstanding:** SEC EDGAR (`dei:EntityCommonStockSharesOutstanding`)
- **Risk-free rates:** FRED (`DGS1`, `SOFR`, `DGS10`) via CSV endpoint

Key features implemented:

- **Dividend adjustment:** $\text{AdjClose_TR} = \text{Close} + \text{CumDividends}$ — cumulative per-share dividends added back to close price to produce a total-return series used as equity value E_t in KMV
- **KMV Debt D:** $\text{CurrentDebt} + 0.5 \times \text{LongTermDebt}$ — computed from SEC GAAP tags with a priority-ordered fallback candidate list per field
- **totalLiabilities fallback:** where SEC `Liabilities` tag is unavailable, computed as $\text{totalAssets} - \text{stockholdersEquity}$ (accounting identity); source method recorded in `_tag` column
- **DELL special case:** multi-class shares (Class C + Class D) summed from SEC filings; hardcoded fallback from 2026-06-09 10-Q (848,171,289 shares total)
- **ORCL special case:** long-term debt filed under `LongTermNotesPayable` — added to `FIELD_DEFS` candidate list
- **Data quality report:** automated check on missing values, negative values, and KMV-ready row count per ticker
- **Output:** per-company CSV files + Excel workbook (6 sheets) + 5 charts; all saved to `./data/<TICKER>/`
- **Team coordination:** hosted the weekly team meeting on Thursday to sync progress across all three pipelines and align on data format for Week 3
- **Group weekly report:** compiled and wrote this weekly report based on individual contributions from all team members

Pipeline steps:

Step	Description
0	Install & import
1	Configuration (tickers, CIK map, output directory)
2	Download 2-year daily price history via Yahoo Chart API
3	Download annual financials + shares from SEC EDGAR
4	Compute daily market cap ($(AdjClose_TR \times Shares)$)
5	Download DGS1, SOFR, DGS10 from FRED
6	Build merged master dataset per company
7	Export to Excel
8	Data quality check
9	File summary

2.2 Haixing Tan — Excel Workbook (`build_financials.mjs`)

Built a JavaScript-based pipeline producing a single consolidated Excel workbook for all 10 companies with full audit trail and data validation.

Data sources used:

- **Company financials:** SEC EDGAR XBRL API — same CIK-based approach, with `fieldDefs` mapping GAAP tags to financial concepts
- **Stock prices:** Yahoo Finance Chart API (direct HTTP, same approach as Shenyuan)
- **Interest rates:** FRED CSV endpoint

Key features implemented:

- **GAAP tag mapping** (`fieldDefs`): priority-ordered candidate lists for `totalDebt`, `currentDebt`, `currentLiabilities`, `longTermDebt`, `longTermLiabilities`, `dividendPerShare`, `dividendCash`, etc. — same methodology adopted by Shenyuan in the notebook
- **DELL special case:** extracted shares from latest 10-Q cover page inline XBRL; summed Class A + Class B + Class C shares
- **Amounts in \$mm:** all dollar figures converted to millions for readability

- **Dividend adjustment factor:** $\text{AdjClose} / \text{Close}$ per day

Excel workbook structure:

Sheet	Contents
Summary	Latest market snapshot: closing price, dividend adjustment factor, shares outstanding, market cap, latest rates
Company Financials	2 fiscal years per company: debt, liabilities, dividends, SEC tags used
Rates	2 years of daily DGS1 and SOFR
Prices	2 years of daily closing prices for all 10 companies
Sources Audit	Source URLs, SEC tags, periods, and notes for reproducibility
Checks	Validation: 10 companies present, 20 company-year rows, no missing market caps, sufficient price/rate observations, no Excel formula errors

Formatting: automated header colors, number/percentage/currency formats, column widths, frozen headers, conditional formatting on validation checks.

2.3 Guanming Lin — Python Downloader (`src/stock_data_downloader.py`)

Built a command-line Python script providing granular per-company file output with full quarterly and annual financial statement coverage.

Data sources used:

- **Stock prices + financials:** `yfinance` Python package
- **Company/ticker resolution:** SEC company ticker list (`sec.gov/files/company_tickers.json`)
- **Interest rates:** FRED CSV (`DGS1`), (`SOFR`)

Key features implemented:

- **CLI interface:** `python src/stock_data_downloader.py COST KO DELL ... --period 2y --interval 1d`
- Quarterly and annual income statement, balance sheet, and cash flow per company
- Dividend adjustment series: `price_history_with_dividend_adjustment.csv`

- Market cap history: closing price × current shares outstanding
- Debt & liability metrics extracted to (debt_liability_metrics_2y.csv)
- Combined output: (data/company_summary.csv), (data/manifest.json), (data/macro_rates/)

Output structure per company ((data/<TICKER>/)):

```
price_history.csv
price_history_with_dividend_adjustment.csv
market_cap_history.csv
quarterly_income_statement.csv / balance_sheet.csv / cash_flow.csv
annual_income_statement_2y.csv / balance_sheet_2y.csv / cash_flow_2y.csv
debt_liability_metrics_2y.csv
summary.json
```

3. Data Results

3.1 Market Snapshot (as of June 24-25, 2026)

Ticker	Company	Close (\$)	Market Cap (\$B)	Shares (M)	Div Rate (\$/yr)	Total Debt (\$B)	Total Liab (\$B)
COST	Costco Wholesale	942.24	417.9	443.5	5.88	10.2	47.9
KO	Coca-Cola	80.42	346.0	4,302.5	2.12	44.6	70.5
DELL	Dell Technologies	409.45	264.6	646.1 ⁺	2.52	31.9	103.8
ORCL	Oracle	152.46	439.2	2,880.5	2.00	167.4	218.7
PNC	PNC Financial	245.28	98.5	401.6	6.80	68.5	512.9
WMT	Walmart	115.78	921.4	7,958.1	0.99	75.5	178.5
INTU	Intuit	255.07	69.8	273.5	4.80	6.9	17.2
AMZN	Amazon	227.01	2,442.0	10,757.1	—	235.5	407.0
T	AT&T	22.42	155.8	6,948.3	1.11	159.8	291.7
KHC	Kraft Heinz	23.47	27.8	1,185.8	1.60	21.1	40.0

† DELL shares: single-class figure from yfinance; Shenyuan's notebook uses 848.2M (Class C + Class D summed from SEC 10-Q)

3.2 Macro Rates (as of June 24, 2026)

Rate	FRED Series	Value
1-Year Treasury CMT	DGS1	3.99%
Secured Overnight Financing Rate	SOFR	3.62%
10-Year Treasury CMT	DGS10	4.41%